

 Marwadi University Marwadi Chandarana Group		Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: Blog Application using Firebase	
Experiment No:- 10	Date:- 20-11-2024	Enrollment No:- 92200133030	

Objective:- Blog Application using Firebase

Code :-

➤ **Lib**

- **models**

- **blog.dart**

```
class Blog {
  String? title;
  String? description;
  String? id;
  Blog({this.title, this.description, this.id});
}
```

- **services**

- **firebase_services.dart**

```
import 'package:cloud_firestore/cloud_firestore.dart';
import '../models/blog.dart';

class FirebaseServices {
  var db = FirebaseFirestore.instance;
  addBlog(String title, String description) {
    // Create a new note with a title and description
    final note = <String, dynamic>{"title": title, "description": description};

    // Add a new document with a generated ID
    db.collection("blog").add(note).then((DocumentReference doc) =>
      // ignore: avoid_print
      print('DocumentSnapshot added with ID: ${doc.id}'));
  }

  Stream<QuerySnapshot> fetchBlog() {
    Stream<QuerySnapshot> collectionStream =
      FirebaseFirestore.instance.collection('blog').snapshots();
    return collectionStream;
  }

  // Stream documentStream =
  FirebaseFirestore.instance.collection('users').doc('ABC123').snapshots();

  updateBlog(Blog blog) {
    return db
      .collection("notes")
```

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```

        .doc(blog.id)
        .update({'title': blog.title, 'description': blog.description})
        // ignore: avoid_print
        .then((value) => print("Note Updated"))
        // ignore: avoid_print
        .catchError((error) => print("Failed to update Note: $error"));
    }

    deleteBlog(id) {
      return db
        .collection("blog")
        .doc(id)
        .delete()
        // ignore: avoid_print
        .then((value) => print("User Deleted"))
        // ignore: avoid_print
        .catchError((error) => print("Failed to delete user: $error"));
    }
  }
}

```

- **screen**

- **blog_details_screen.dart**

```

import 'package:flutter/material.dart';

import '../models/blog.dart';
import '../services/firebase_services.dart';

class BlogDetailsScreen extends StatefulWidget {
  final Blog? blog;
  const BlogDetailsScreen({super.key, this.blog});
  @override
  State<BlogDetailsScreen> createState() => _BlogDetailsScreenState();
}

class _BlogDetailsScreenState extends State<BlogDetailsScreen> {
  final _formKey = GlobalKey<FormState>();
  late String _title;
  late String _content;

  @override
  void initState() {
    super.initState();
    if (widget.blog != null) {
      _title = widget.blog!.title!;
      _content = widget.blog!.description!;
    } else {
      _title = '';
      _content = '';
    }
  }
}

```

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```

    }

    void _saveBlog() {
      if (_formKey.currentState!.validate()) {
        _formKey.currentState!.save();
        // final blog = Blog(
        //   id: widget.blog?.id,
        //   title: _title,
        //   description: _content,
        // );
        //todo code to store on firestore
        FirebaseServices().addBlog(_title, _content);
        Navigator.pop(context);
      }
    }

    @override
    Widget build(BuildContext context) {
      return Scaffold(
        appBar: AppBar(
          title: Text(widget.blog == null ? 'New Blog' : 'Edit Blog'),
        ),
        body: Padding(
          padding: const EdgeInsets.all(16.0),
          child: Form(
            key: _formKey,
            child: Column(
              children: [
                TextFormField(
                  initialValue: _title,
                  decoration: const InputDecoration(labelText: 'Title'),
                  validator: (value) {
                    if (value == null || value.isEmpty) {
                      return 'Please enter a title';
                    }
                    return null;
                  },
                  onSave: (value) {
                    _title = value!;
                  },
                ),
                const SizedBox(height: 16.0),
                TextFormField(
                  initialValue: _content,
                  decoration: const InputDecoration(labelText: 'Content'),
                  validator: (value) {
                    if (value == null || value.isEmpty) {
                      return 'Please enter content';
                    }
                    return null;
                  },
                  onSave: (value) {

```

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```

        _content = value!;
      },
    ),
    const SizedBox(height: 16.0),
    ElevatedButton(
      onPressed: _saveBlog,
      child: const Text('Save'),
    ),
  ],
),
),
),
);
}
}

```

○ **blog_list_screen.dart**

```

import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:flutter/material.dart';
import 'package:blog_app/services/firebase_services.dart';

import '../models/blog.dart';
import 'blog_details_screen.dart';

class BlogListScreen extends StatelessWidget {
  const BlogListScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Blog'),
      ),
      body: StreamBuilder<QuerySnapshot>(
        stream: FirebaseServices().fetchBlog(),
        builder: (BuildContext context, AsyncSnapshot<QuerySnapshot> snapshot) {
          if (snapshot.hasError) {
            return const Text('Something went wrong');
          }

          if (snapshot.connectionState == ConnectionState.waiting) {
            return const Center(child: CircularProgressIndicator());
          }

          if (!snapshot.hasData ||
              snapshot.data == null ||
              snapshot.data!.docs.isEmpty) {
            return const Text("No Blog found");
          }
        },
      ),
    );
  }
}

```

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```

    }

    return ListView(
      children: snapshot.data!.docs.map((DocumentSnapshot document) {
        Map<String, dynamic>? data =
          document.data() as Map<String, dynamic>;

        if (data == null) {
          return const ListTile(
            title: Text('No Data'),
          );
        }

        // Safely handle null for title and description
        String title = data['title'] ?? 'Untitled';
        String description = data['description'] ?? 'No description';

        Blog blog = Blog(
          id: document.id,
          title: title,
          description: description,
        );

        return ListTile(
          title: Text(blog.title!),
          subtitle: Text(blog.description!),
          trailing: Row(
            mainAxisAlignment: MainAxisAlignment.min,
            children: [
              IconButton(
                icon: const Icon(Icons.edit, color: Colors.blue),
                onPressed: () {
                  // Navigate to edit blog screen
                  Navigator.of(context).push(MaterialPageRoute(
                    builder: (context) => BlogDetailsScreen(blog: blog),
                  ));
                },
              ),
              IconButton(
                icon: const Icon(Icons.delete, color: Colors.red),
                onPressed: () {
                  // Confirm deletion before deleting
                  showDialog(
                    context: context,
                    builder: (BuildContext context) {
                      return AlertDialog(
                        title: const Text("Delete Blog"),
                        content: const Text(
                          "Are you sure you want to delete this blog?"
                        ),
                        actions: [
                          TextButton(
                            onPressed: () => Navigator.of(context).pop(),

```

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```

        child: const Text("Cancel"),
      ),
      TextButton(
        onPressed: () {
          FirebaseServices().deleteBlog(document.id);
          Navigator.of(context).pop();
        },
        child: const Text("Delete"),
      ),
    ],
  ),
);
},
);
},
),
],
),
);
}).toList(),
);
},
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Navigator.of(context).push(MaterialPageRoute(
      builder: (context) => const BlogDetailsScreen()));
  },
  child: const Icon(Icons.add),
),
);
}
}

```

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Output:-

